Task-6

Linux uses two distinct execution modes, Kernel Mode & User Mode, each with different privileges that determine how processes interact with system resources. These modes help ensure system stability, security & efficient management of resources.

Kernel Mode :- It is the most privileged execution mode. The kernel, which is the core part of the operating system, runs in this mode & has unrestricted access to system resources like hardware, memory & CPU instructions. It can perform tasks like process management, memory allocation & hardware control.

- Operations in Kernel Mode:
  - Memory management
  - Hardware access (eg., disk, network)
  - Process Management (eg., scheduling, termination)
  - Handling system calls from user-mode processes.

Example: Device drivers interact with hardware & perform operations like disk read/write in kernel mode.

User Mode :- User mode is where most applications run. It has limited access to system resources and cannot directly interact with hardware or critical system structures. Applications must use system calls to request services from the kernel.

- Operations in user mode:
  - Application execution (e.g., text editors, web browsers)
  - Access to system calls (e.g., file handling)
  - Limited memory access (isolated address spaces)

Example :- A text editor can read/write files but cannot modify system files without intervention.

Transition Between Modes :-
User-mode programs interact with the kernel via system calls. when a system call is made, the CPU switches to kernel mode to perform the task & then returns to user mode.

Security Implications :-
The separation between these modes is vital for security. It prevents user-space processes from corrupting system memory or compromising hardware. Only authorized processes, like system services, can escalate to kernel mode, ensuring system integrity.